

IoT-Based Smart Security System on a Door Lock Application



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Abstract The Internet of Things (IoT) is the term that describes the communication between physical object also known as the “things” with the interconnection network. Nowadays, IoT plays a major role in every aspects of lives, i.e., simply communicating with other devices to security of homes, and the role of IoT is seen everywhere. From smart cities to smart home along with medical system, researchers are keen on doing research in these fields. Whatever physical objects we can see it can be transformed into IoT devices with proper data connection and using some tools. It can be controlled and communicate over Internet. Smart bulb to smart dustbin, smart refrigerator, smart mirror, and so on these lists can never end. The keys to a smart home depend upon the security of the door. If the entrance of the home is secured, then the home will be more secure. Nowadays, everyone is living away from his/her home for most of the time because of work, academics, and so many things, but our most expensive and precious things are at home. So many recent security issues are happening from theft to unwanted entry of some persons. Senior citizen and children are there at home, and their security should not be hampered at all when the owner is absent in his home. From the concept of smart home, we a smart door unlocking system has been proposed that uses infrared sensors, arduino board to give alarm when any suspicious activity is going on. Although there are many researches have been done in this smart door unlock system, but our proposed method yields the best result with low cost. Here, user will notify about all the suspicious things when that person will absent also no unauthorized entry will be there, so it leads to better and more secure environment in his home. This paper revolves around the IoT security which mainly focuses on home automation. The proposed system is designed to help an individual secure his/her house from theft; if someone break the door or entered wrong password for the door, it directly informed to the concerned house owner. Home automation provides the automatic control over the appliances at home and secures the home from an intruder.

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1 Introduction

These days, innovation is a fundamental piece of everybody's lives. It impacts a few aspects of regular day to day existence and permits improved social cooperative energy, simple transportation, the ability of enjoying amusement, and media and aides in the progression in medication. The creation of a few gadgets like PDAs and PCs has made a few people dependent on innovation for interfacing with companions, store and recover data like pictures, recordings, archives, and music. The Internet is a typical interface that few gadgets use so as to make the day-by-day life of numerous individuals. Web has assumed a spearheading job in giving quick answers for different issues and has given the capacity and has associated all the remote spots which has added to critical decrease in cost and furthermore vitality utilization. Incident like theft is more common, what is uncommon that people are unaware of advanced systems of intrusion detection. Home mechanization or shrewd home is characterized as inception of innovation inside the home environmental factors to give simplicity and security to its occupants. The innovation of the Internet of Things is utilized to analyze and execute home mechanization. IoT likewise extraordinarily adds to supply the executives and recognition easily of control. The World Wide Web is extraordinarily utilized in home mechanization that gives choices by means of moderate utilization of vitality. Wireless home automation using IoT uses the appliances and the mobile device or the password modules to control the mishappenings at home. Also saves the energy and keeps the home secure. Installing different components is very difficult to maintain, so the proposed system can turn this difficulty into simple and possible. Arduino is the main part used in this system, and with it some sensors, buzzer is used in intrusion detection. Existing system does not have any buzzer system. It only checks for the password; if it matches, then the door opens, and if not, it may get block or remains unopened. But our system uses the buzzer and informing the owner a wrong person is going to enter the house.

2 Literature Survey

On the off chance that there is a message sent on the MQTT broker, at that point entryway lock will get the message and search for what request which has been gotten. The info time for the entryway to open is introduced to a certain time span in the event that the order is an open entryway [1]. The proposed approach is acknowledged utilizing Bluetooth innovation for the network, android-based advanced mobile phone app for the end client access, and client certifications [2]. The ZigBee module in smart door lock works as an interface between sensor centers and the control

module. ZigBee module is actualized in the advanced locking system which goes about as the primary regulator for the general smart home system in the enterprise. The electronic lock is made out of a guideline processor, a ZigBee module, a lock regulator, a

CDMA module, and a camera module [3]. This paper aims to propose a security door lock system based on Raspberry pi technology. Cameras, keypad, and pi-lids are being utilized to provide an alarming system [4]. In [5], a framework has structured that contains sensors to distinguish snag, contact, heat, smoke, and sound. The entire framework is constrained by a PIC microcontroller 16F76. This framework requires additional equipment segments like sensors and GSM modem [6]. The framework utilizes the ZigBee module in advanced entryway lock, and the entryway locks go about as a focal principle controller [7]. A framework has proposed by the author in paper [8] dependent on the RFID innovation which gives a touch LCD screen. Another framework comprises of a form in NFC capacities of a PDA [9]. Here, the author gives emphasis to the essential host has a period breaking point to answer whether to open the entryway or not [10]. The cell phone can get caution buzz when there is a potential gatecrasher and can bolt/open the entryway. The attractive sensor is additionally used to naturally bolt the entryway when it is shut [11–13]. For a smart room automation and localization system, a mobile robot can be used while overcoming challenges associated with localization, perception, cognition, and control while mobile robot can extract data from the sensor and will decide the trajectory position and how to act according to the environment [14]. Here, the author proposed a model which used digital signature for authentication and triple DES encryption standard for data transmission which are implemented using MD5 and RSA algorithm for the design of smart cities [15]. The reinforcement learning techniques and multi-objective ant colony optimization (MOACO) algorithms have been applied to deal with the accurate resource allocation between the end users. Here, swarm intelligence-based and reinforcement learning technique used here with neural coaching for task execution [16–18].

This research paper is organized as followed: In Sect. 1, an introduction is given; Sect. 2 deals with some literature survey. In Sect. 3 method, material and proposed models have been discussed as well as present experimental results and discussions with proper explanations. In Sect. 4, finally, we concluded the research work with future scope

3 Method and Material

Some of the features of proposed system are digital code lock venture—is a straightforward electronic number lock framework or an electronic blend lock utilizing Arduino—which has a preset 4-digitsecret phrase put away inside the program. The framework gathers 4-digitclient input by using the password module, contrasts the client input and the preset secret key inside program, and if the client input and put away secret key matches, access will be allowed (by opening the entryway with the

assistance of servo motor) and deactivating the IR sensor which is installed at the door. On the off chance that there is a confuse between client input and put away secret word, access will be denied (by not opening the shut door), and if wrong password is entered three times consecutively, then buzzer will ON, and if the person enters the room without entering the password, then also buzzer will ON, as the IR sensor is already in activation mode.

3.1 *Components Required*

- **Arduino Uno:** It is an open hardware stage when users do his program on using it.
- **LCD Display:** Liquid crystal display (LCD) is a display which surface is flat in nature it works on the concept of light modulating properties with the help of polarizer to produce light beam.
- **Keypad:** allows an individual to easily enter numeric values into a computer
- **IR Sensor:** An infrared sensor is an electronic gadget that measures infrared lighting from objects. It plays a vital role in security application. It senses the object from its surroundings and detects the motion of it.
- **Bread Board:** A breadboard is a rectangular plastic board with a lot of minuscule openings or holes in it. These holes let the user effectively embed electronic parts to model (which means to fabricate and test an early form of) an electronic circuit
- **Buzzer:** A bell or beeper is a sound flagging gadget
- **Servo Motor:** A servo motor works on the closed-loop mechanism where it gives positional feedback. This motor is controlled by an outer electric signal.
- **Potentiometer:** A potentiometer is a three-terminal resistor with a sliding or turning contact that frames a customizable voltage divider. It can be defined as an extension to resistor but it is having a third terminal on it where resistor has only two terminals. Here, the third terminals can be adjusted to give any fraction of voltage to the end of the resistor.
- **220- Ω Resistor:** Resistor can be simply defined as a device which has the ability to resist the passage of current. Resistors are used to control the current flow adjusting the levels, and it is estimated in units of ohms (Ω),
- **Connecting Wires:** used to expand the terminating line or leg wires in an electric impacting circuit.
- **USB Cable:** Used to connect Arduino and the system.

3.2 *Flow Chart of Proposed Work*

A flow chart diagram of the smart security system on door lock application is represented in Fig 1; here, first user inputs the password from the keypad ('I' is a variable which is set to 0) and that is matched with the preset password; if it is matched, the

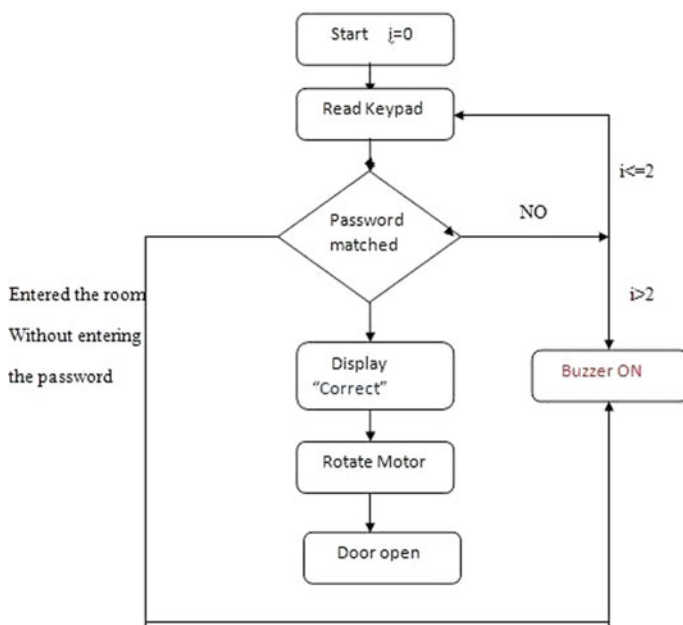


Fig. 1 Flow chart of proposed work

LCD will display “Correct,” and the servo motor will rotate and door will be opened (IR sensor which is fitted near the door will be de-activated). If the user inputs the wrong password for the first time, then i will be incremented by 1, and if repeats, then i value will be incremented, and if $i > 2$, then BUZZER will be ON. If the user enters the room without entering the password (i.e., by breaking the lock or door), then IR sensor which is already in activated mode leads the BUZZER to be ON.

3.3 Implementation

Custom key will collect the value from keypad using `getkey()` method and that value is stored in `value[]`. Count variable is used in order to count the number of attempts of the user to enter the password. The present value or the password is “98BC” and that is matched with the `value[]`, and if matched door will be opened and LCD shows “correct,” serial monitor shows “Lock open” and door gets opened else serial monitor shows “password incorrect” and door remains closed. In this if k , i.e., count > 2 OR $\text{data_ir} = 1$ i.e., IR sensor senses the presence (when password is Incorrect) then buzzer will turn ON. If the password is correct, flag value will be set to 1 else 0. If the person enters the room without entering password, flag will be 0, and IR sensor senses the presence then buzzer will be ON.

3.4 Final Setup and Working Principle

A block diagram of smart security system is given in Fig 2; here, we connected the keypad to the seven digital pins and LCD pins to the analog pins of the Arduino. Servo motor has three pins in which red is connected to +5v, brown is connected to ground, and the yellow is connected to pin no. 9 of Arduino. Buzzer's one pin is connected to +5v, and other is connected to pin 11. IR sensor's one pin is connected to +5v, and other is connected to pin 12.

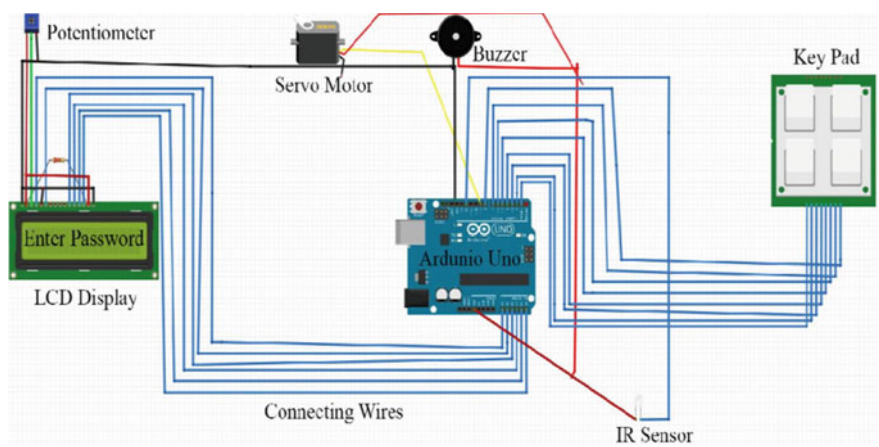


Fig. 2 Block diagram of smart security system

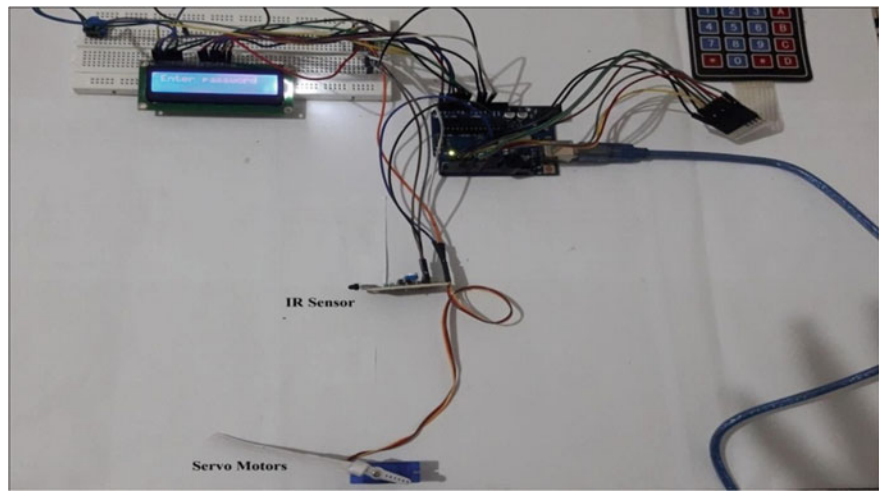


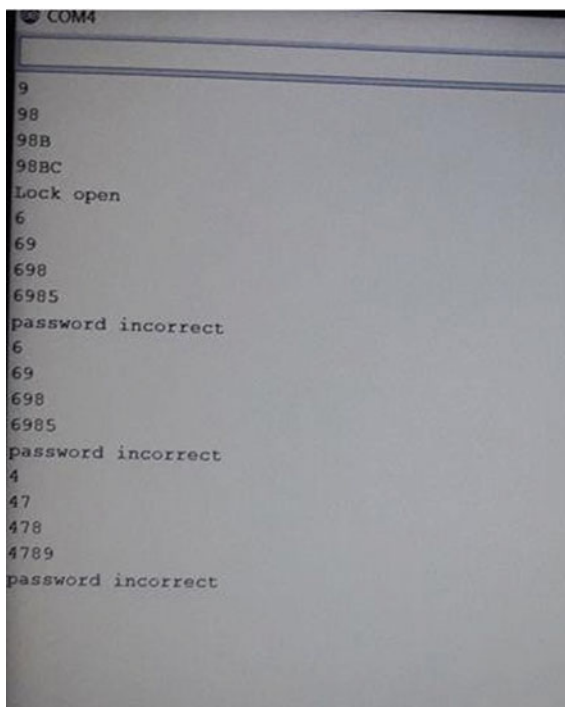
Fig. 3 Digital code lock venture

Digital code lock venture system is represented in Fig. 3, and user enters the password which is matched with the preset password in the Arduino, and if it matches, servo motor, which is connected to Arduino, will be activated, and the motor rotates thus opens the door. And if the password inputted is incorrect, then no action will take place, and if it is repeated three times, then Arduino activates the Buzzer and it will be ON. If the user enters the room without entering the password IR sensor, which is already in activation mode, senses the presence of object thus in turn buzzer will be ON (if password is correct, IR sensor will be deactivated).

3.5 Result in the Serial Monitor

A demonstrating result on serial monitor is represented in Fig 4, while the user enters the password digit by digit is printed in the serial monitor and if the password matches, it will print “Lock Open,” and if the password is incorrect, it will display “Password Incorrect.” The user will also be getting the message in the LCD display while the user enters the password, and if matches, it will display as “Correct,” if it is wrong, it will display as “Enter Password.”

Fig. 4 Demonstrating results on serial monitor



4 Conclusion and Future Scope

In today's generation, everyone is dependent on Internet either directly or indirectly. This paper consists of two parts hardware and software. Software part defines the algorithm that we built for the intrusion detection. Hardware part describes how the system was built and the modules we used. The proposed system provides a mini computer-based home security system with a low cost and stable operating system. Main advantage is that it is fully automated, once installed does not require any kind of human interaction. In future works, it tends to be wanted to utilize the sustainable power source and subsequently can be remote. And it can also take the photos of the users who had inputted the wrong password and send it to the respective owner.

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